

History of Computers and Moore's Law

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This presentation contains material that is

These slides are based on the book:

Patterson, David. "Computer organization and design RISC-V edition: the hardware." (2017).

History of Computers: The Abacus

1. Shu'Tien Li (1958) gives a detailed description of the Chinese abacus evolution which dates back according to him to 110 B.C.
2. Some others claim that the abacus is a Roman/Greek invention (~500 B.C.), but what seems is that the Romans and Greeks have rather different formats which utilized the symbolic system to calculate invented by the Chinese.
3. Other authors claim the abacus was invented in Mesopotamia some 5000 years ago.

History of Computers

Humans are lazy...



Chaplin's Modern Times

<https://tenor.com/en-CA/view/comedy-classic-modern-times-charlie-chaplin-funny-gif-3309841>

History of Computers: Automata (not in book!!)

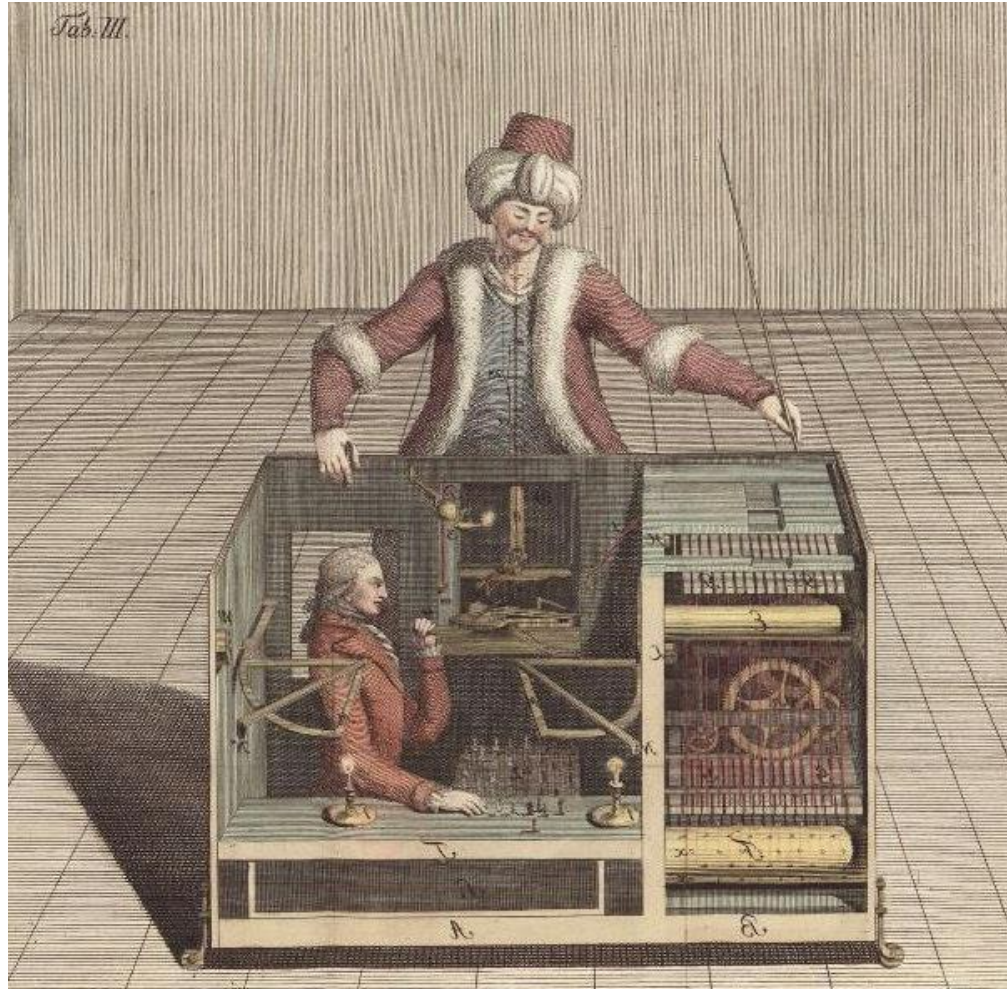
[Antikythera mechanism](#) from 150 to 100 BC



History of Computers: Automata (not in book!!)



History of Computers: Automata (not in book!!)



Coloured engraving from Joseph Racknitz's 1789 pamphlet which attempted to reveal the secret workings of William Kempelen's alleged chess-playing automaton "The Turk"

History of Computers: Automata (not in book!!)

Check:

<https://automata.falculture.org/a-brief-history-of-automata-and-mechanical-toys/>

History of Computers

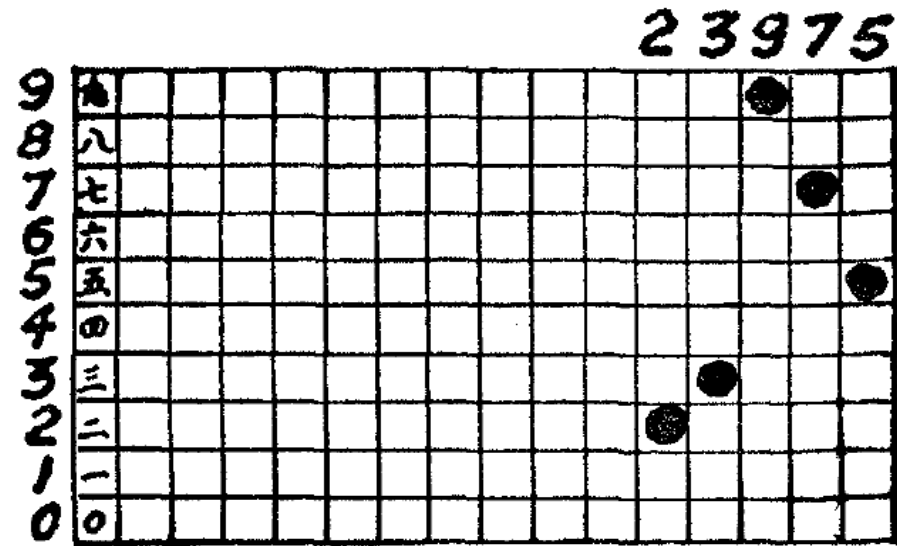
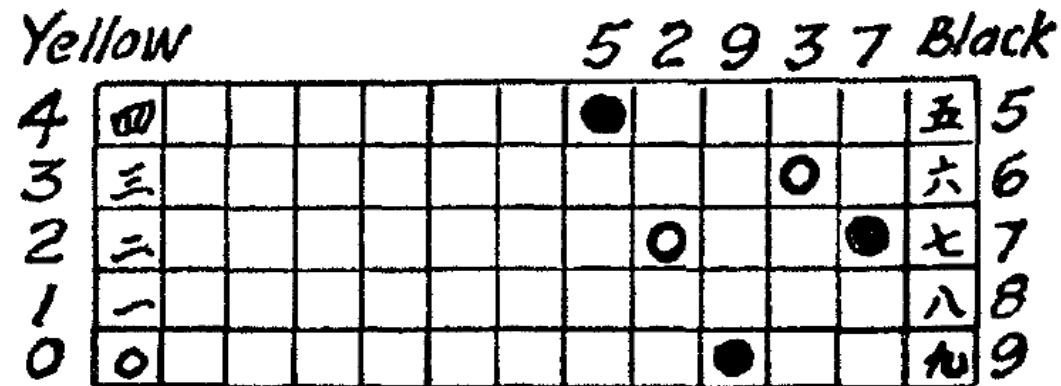
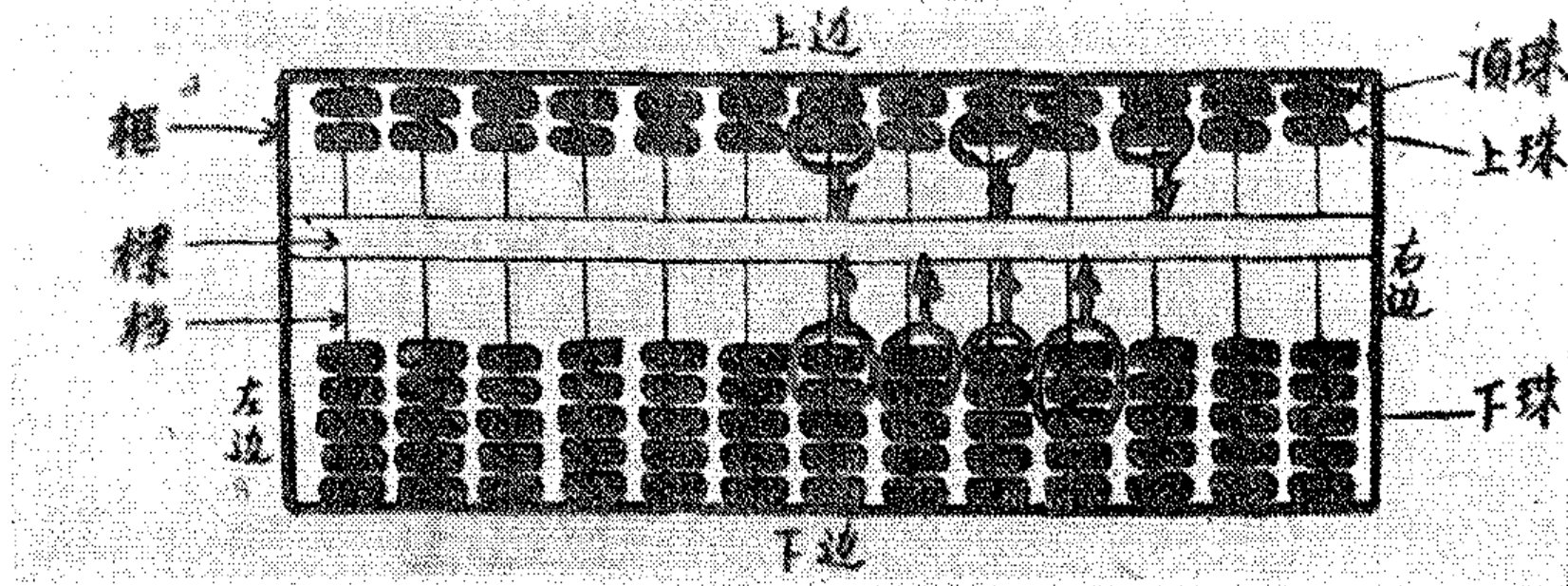


FIG. 1. Chu Pan, or Bead Tray



Li, Shu-T'ien. "Origin and development of the Chinese abacus."
Journal of the ACM (JACM) 6.1 (1959): 102-110.

History of Computers



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History of Computers

1642: Blaise Pascal



https://en.wikipedia.org/wiki/Pascal%27s_calculator#/media/File:17th-century-mechanical-calculators_-_Detail.jpg

History of Computers

1801: Joseph Marie Jacquard's loom: punch cards



Jacquard loom in the Making It gallery in the National Museum of Scotland.

<https://www.nms.ac.uk/explore-our-collections/stories/science-and-technology/jacquard-loom/>

History of Computers

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History of Computers

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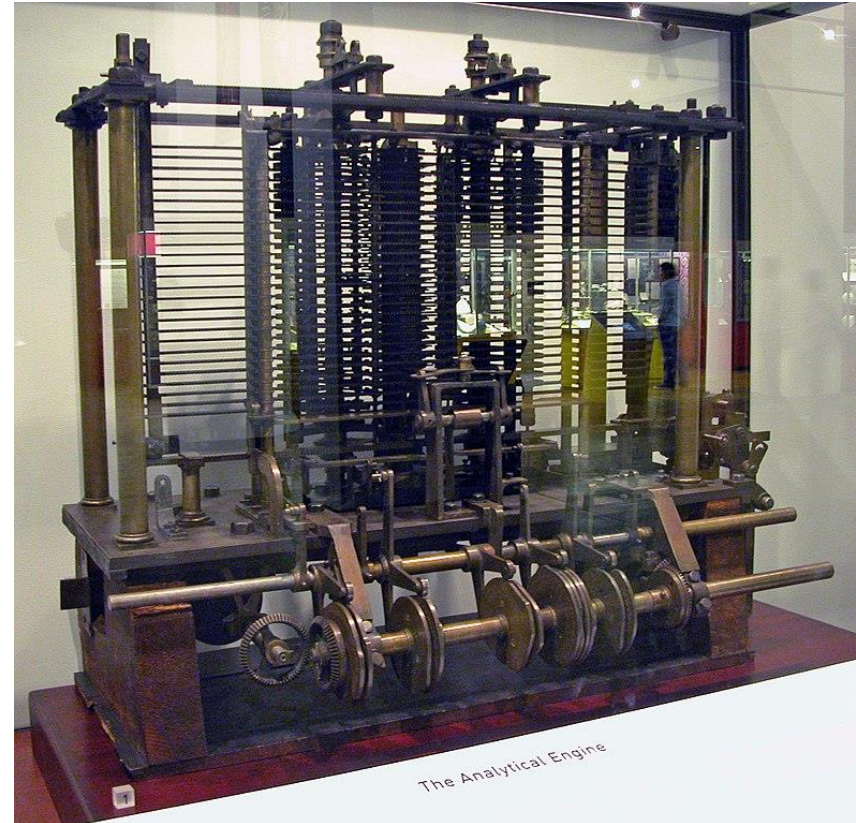
https://en.wikipedia.org/wiki/Jacquard_machine#/media/File:A_Jacquard_loom_showing_information_punchcards,_National_Museum_of_Scotland.jpg

<https://www.nms.ac.uk/explore-our-collections/stories/science-and-technology/jacquard-loom/>

History of Computers

1800's: Charles Babbage and Augusta Ada Lovelace.

George Boole's Binary (Boolean) logic



Analytical Engine, trial model, Science Museum Group

<https://blogs.bodleian.ox.ac.uk/adalovelace/2018/07/26/ada-lovelace-and-the-analytical-engine/>

History of Computers

George Boole: Binary logic

[Stanford Encyclopedia of Philosophy](#)

x	y	$1 - x$	$x - x^2$	xy	$x + y$	$x - y$	$x + y - xy$	$x + y - 2xy$
1	1	0	0	1	2	0	1	0
1	0	0	0	0	1	1	1	1
0	1	1	0	0	1	-1	1	1
0	0	1	0	0	0	0	0	0

<https://plato.stanford.edu/entries/boole/#ContBackBoolWorkLogi>

History of Computers

George Boole: Binary logic

[Stanford Encyclopedia of Philosophy](https://plato.stanford.edu/entries/boole/)

Secondary Propositions	<i>MAL</i> (1847)		<i>LT</i> (1854)	
X is true	$x = 1$	p.51	$x = 1$	p.172
X is false	$x = 0$	p.51	$x = 0$	p.172
X is true and Y is true	$xy = 1$	p.51	$xy = 1$	p.172
X is true or Y is true (inclusive)	$x + y - xy = 1$	p.52	—	
X is true or Y is true (exclusive)	$x - 2xy + y = 1$	p.53	$x(1 - y) + y(1 - x) = 1$	p.173
If X is true then Y is true	$x(1 - y) = 0$	p.54	$x = vy$	p.173

<https://plato.stanford.edu/entries/boole/#ContBackBoolWorkLogi>

History of Computers

George Boole: Binary logic: AND operator $p * q / p \wedge q$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

If p/q is TRUE (T)=1. If p/q is FALSE (F)=0

https://en.wikipedia.org/wiki/Truth_table

History of Computers

George Boole: Binary logic: OR operator

$$p+q/p \vee q$$

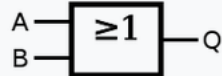
<i>p</i>	<i>q</i>	<i>p ∨ q</i>
T	T	T
T	F	T
F	T	T
F	F	F

If p/q is TRUE (T)=1. If p/q is FALSE (F)=0

https://en.wikipedia.org/wiki/Truth_table

History of Computers

Logic gates: Binary logic

AND			$A \cdot B$ or $A \wedge B$	<table><tr><th colspan="2">Input</th><th>Output</th></tr><tr><th>A</th><th>B</th><th>Q</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	Input		Output	A	B	Q	0	0	0	0	1	0	1	0	0	1	1	1
				Input		Output																
				A	B	Q																
				0	0	0																
				0	1	0																
1	0	0																				
1	1	1																				
OR			$A + B$ or $A \vee B$	<table><tr><th colspan="2">Input</th><th>Output</th></tr><tr><th>A</th><th>B</th><th>Q</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	Input		Output	A	B	Q	0	0	0	0	1	1	1	0	1	1	1	1
				Input		Output																
				A	B	Q																
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History of Computers

1904: The vacuum tube [John Ambrose Fleming](#)

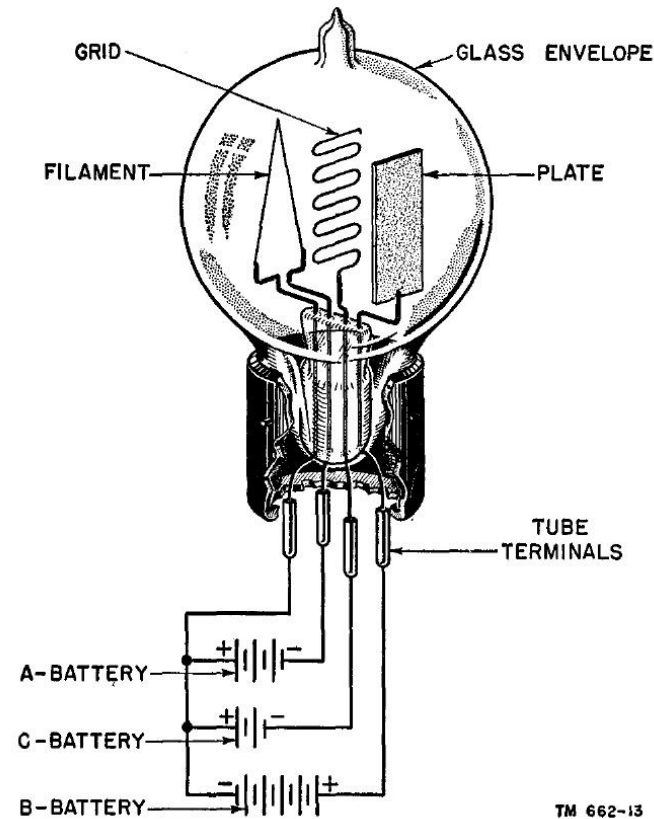
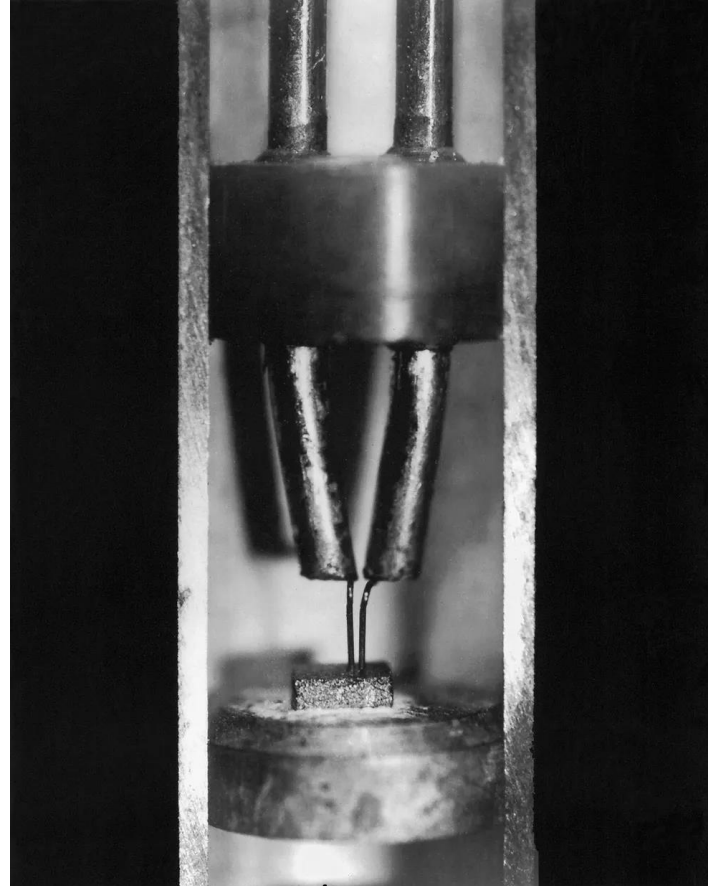


Figure 4. Construction of DeForest's three-element tube, or triode.

https://en.wikipedia.org/wiki/Vacuum_tube#/media/File:TRIO DE_TM11_662_FIG_4.jpg

History of Computers

November 1947 The Transistor: Bardeen and Brattain

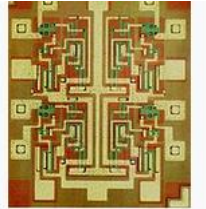
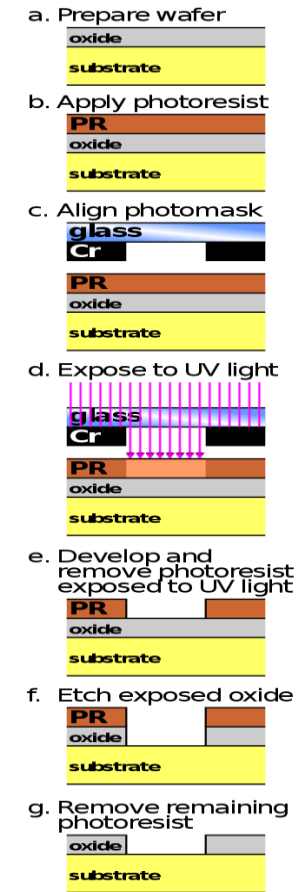


<https://spectrum.ieee.org/transistor-history>

https://www.google.com/search?client=firefox-b-d&sca_esv=564785002&q=cartoon+from+1940%27s+of+vacuum+tube&tbm=vid&source=lnms&sa=X&ved=2ahUKEwjk tPjJ7qWBAXgAzQIHW8oBOIQ0pQJegQICRAB&biw=1408&bih=653&dpr=1.36#fpstate=ive&vld=cid:c7906811,vid:sxxVDXliz9Q,st:0

History of Computers

Microchip: **Complementary metal–oxide–semiconductor (CMOS, pronounced "sea-moss")**



MOSFET scaling
(process nodes)

10 μm – 1971
6 μm – 1974
3 μm – 1977
1.5 μm – 1981
1 μm – 1984
800 nm – 1987
600 nm – 1990
350 nm – 1993
250 nm – 1996
180 nm – 1999
130 nm – 2001
90 nm – 2003
65 nm – 2005
45 nm – 2007
32 nm – 2009
22 nm – 2012
14 nm – 2014
10 nm – 2016
7 nm – 2018
5 nm – 2020
3 nm – 2022

Future

metal-oxide-semiconductor field-effect transistor (MOSFET, MOS-FET, or MOS FET)

Moore's Law

Year	Technology used in computers	Relative performance/unit cost
1951	Vacuum tube	1
1965	Transistor	35
1975	Integrated circuit	900
1995	Very large-scale integrated circuit	2,400,000
2020	Ultra large-scale integrated circuit	500,000,000,000

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Moore's Law

$$\text{Time} = \text{Seconds/program} \times (\text{clock cycles/instruction}) \times (\text{Seconds/clock cycle})$$

Application Software	System software & Hardware architecture	Chip design and Semiconductor Material
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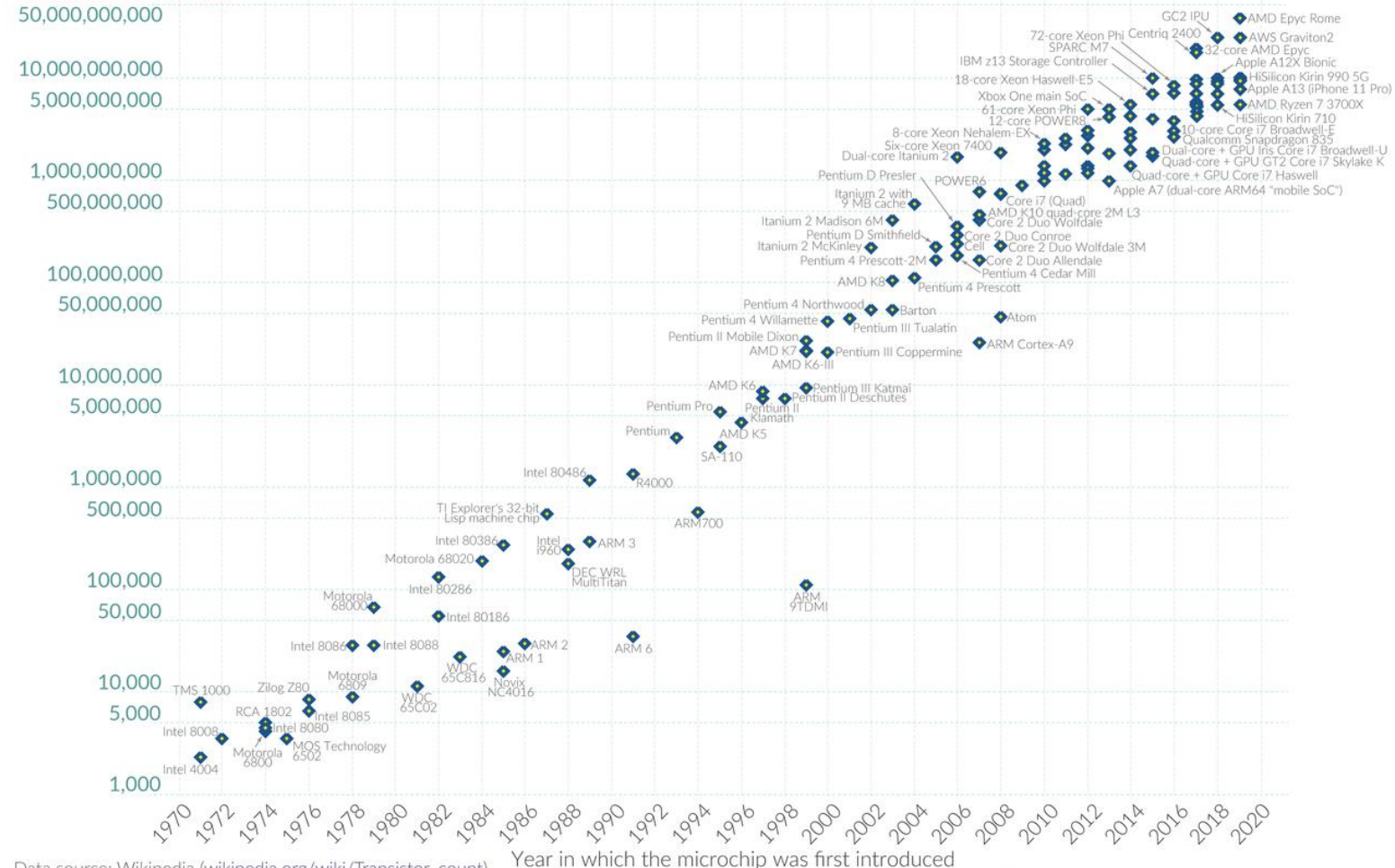
Moore's Law

Moore's Law: The number of transistors on microchips doubles every two years

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

Our World
in Data

Transistor count



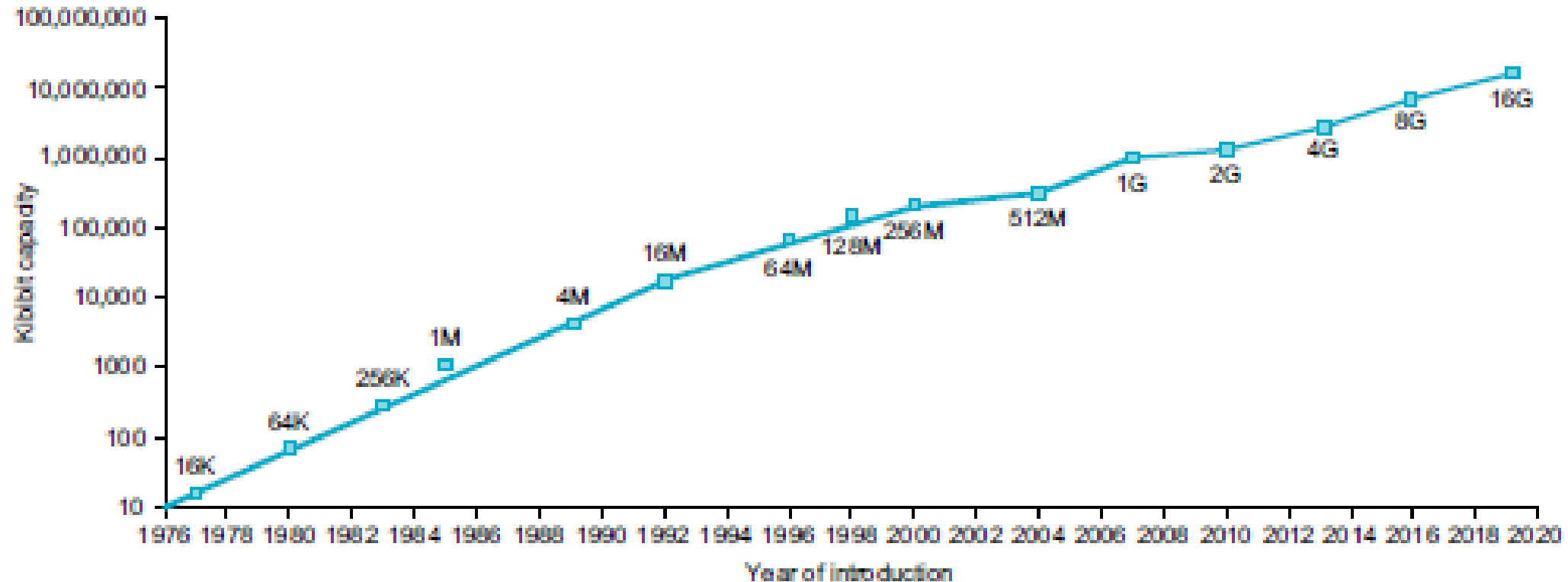
Data source: Wikipedia ([wikipedia.org/wiki/Transistor_count](https://en.wikipedia.org/wiki/Transistor_count))

OurWorldinData.org – Research and data to make progress against the world's largest problems.

Licensed under CC-BY by the authors Hannah Ritchie and Max Roser.

https://en.wikipedia.org/wiki/Moore%27s_law#/media/File:Moore's_Law_Transistor_Count_1970-2020.png

Moore's Law



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